

Preliminary Linear Algebra and Related Calculations

1. Simple Matrix Operations

```
A= [  
    1    4  
    2    5  
    3    6];
```

```
sA=size(A); % order of a matrix – rows x columns
```

```
B=A';
```

```
% Transpose of the matrix A (interchange rows and columns  
% That is rows becomes columns and columns become rows
```

```
% B =  
%  
%    1    2    3  
%    4    5    6
```

```
sB = size(B);  
C=A*A';  
sC=size(C);
```

```
% C =  
%  
%    17    22    27  
%    22    29    36  
%    27    36    45
```

```
D=A'*A;
```

```
% D =  
%  
%    14    32  
%    32    77    (Note Matrix is symmetric, D=D')
```

```
out =[sA sB sC];
```

```
% out =
```

```
%
```

```
%      3 x 2      2 x 3      3 x 3
```

```
M = [
```

```
      1      2
```

```
      3      4 ];
```

```
DeterminantM =      -2
```

```
>> InvM % Inverse of a matrix
```

```
InvM =
```

```
      -2.0000      1.0000
```

```
      1.5000     -0.5000
```

```
InvM =
```

```
      -2.0000      1.0000
```

```
      1.5000     -0.5000
```

```
>> Check if M*InvM=Identity Matrix
```

```
      1.0000      0
```

```
      0.0000      1.0000
```

% Find eigenvalues and eigenvectors of Matrix D

% D =

%

% 14 32

% 32 77

[eve eva]=eig(D); % eva = eigenvalue eve = eigenvector

% eva =

%

% 0.5973 0

% 0 90.4027

% eve =

%

% -0.9224 0.3863

% 0.3863 0.9224

2. Detailed Calculations for the Inverse

a) Finding Determinant of a [2x2] matrix

$$\text{Let } \mathbf{M} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\text{Det}(\mathbf{M}) = 1 \cdot 4 - 3 \cdot 2 = -2$$

b) Find the inverse of M i.e., $\mathbf{M}^{-1}$

$$\mathbf{A}^{-1} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^{-1}$$

$$\mathbf{A}^{-1} = \left(\frac{1}{1 \cdot 4 - 3 \cdot 2} \right) \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

Procedure:

- I.** Find $\text{Det}(\mathbf{A})$
- II.** Interchange diagonal elements of **A** and change the signs of the remaining elements
- III.** Check if **$\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$**

$$\mathbf{A}\mathbf{A}^{-1} = \left(\frac{1}{1 \cdot 4 - 3 \cdot 2} \right) \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

$$\mathbf{A}\mathbf{A}^{-1} = \left(-\frac{1}{2} \right) \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}$$

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The Identity Matrix

3. Detailed Calculations for the Eigenvalues and Eigenvectors

1. Show the steps for eigenvalues calculation for matrix D

$$\mathbf{D} = \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix}$$

To find eigenvalues, we find the roots of the characteristic equation which is obtained as follows

$$\det(\mathbf{D} - \lambda \mathbf{I}) = \det \left(\begin{bmatrix} 14 - \lambda & 32 \\ 32 & 77 - \lambda \end{bmatrix} \right)$$

$$D = (14 - \lambda)(77 - \lambda) - 32^2 = 0$$

Solving this quadratic equation we get

$$(14 * 77 - 91 + \lambda^2) - 1024 = 0$$

$$\lambda^2 - 91 + 54 = 0$$

$$= \frac{91 \pm \sqrt{91^2 - 4 * 54}}{2}$$

$$\lambda_1 = \frac{91.000 + 89.805}{2} = 90.403$$

$$\lambda_2 = \frac{91.000 - 89.805}{2} = 0.5973$$

Now to find eigenvectors

$$Dx = \lambda x$$

$$\begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 90.403 * \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0.5973 * \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

After expanding the above two equations we land up with the following expressions

$$14 x_1 + 32 x_2 = 90.403 x_1$$

$$14 x_1 + 32 x_2 = 0.5973 x_1$$

From the above, we get two equations relating x_1 with x_2

$$32 x_2 = 76.403 x_1 \quad (1)$$

$$32 x_2 = -13.403 x_1 \quad (2)$$

Letting $x_1 = 1$ we get $x_2 = 2.388$ from (1) and

Letting $x_1 = 1$ we get $x_2 = -0.419$ from (2)

The two eigenvectors are

$$v_1 = \begin{bmatrix} 1 \\ 2.388 \end{bmatrix} \text{ and } v_2 = \begin{bmatrix} 1 \\ -0.419 \end{bmatrix}$$

Usually the resulting eigenvectors are scaled by dividing them with their respective magnitudes

$$v_1 = \begin{bmatrix} 1 / \sqrt{1^2 + 2.388^2} \\ 2.388 / \sqrt{1^2 + 2.388^2} \end{bmatrix} \text{ and } v_2 = \begin{bmatrix} 1 / \sqrt{1^2 + (-0.419)^2} \\ -0.419 / \sqrt{1^2 + (-0.419)^2} \end{bmatrix}$$

$$v_1 = \begin{bmatrix} 0.3863 \\ 0.9223 \end{bmatrix} \text{ and } v_2 = \begin{bmatrix} 0.9223 \\ -0.3863 \end{bmatrix}$$

Matlab code given below, you may check with python code

```
% D =
%
% 14 32
% 32 77

[eve eva]=eig(D); % eva = eigenvalue eve = eigenvector

% eva =
%
% 0.5973 0
% 0 90.4027

% eve =
%
% -0.9224 0.3863
% 0.3863 0.9224
```

4. Geometric interpretation of eigenvalues and eigenvectors

Suppose there is a vector, and it is acted upon by a square matrix. What happens to the vector is the point of interest. Usually, when a matrix **M** acts on a vector **v**, both the direction and magnitude of **v** may change.

However in special cases, only the magnitude changes for the vector but no change in direction

v will experience one of the following

- a) increase $\lambda > 1$
- b) decrease $\lambda < 1$

c) Vanishes/collapses into a zero vector (no magnitude or direction)

$$\lambda=0,$$

d) reverse in direction $\lambda<0$.

In such a case, $\mathbf{v}$ is termed as the eigenvector and the amount by which $\mathbf{v}$ is expanded or contracted is called as the eigenvalue.

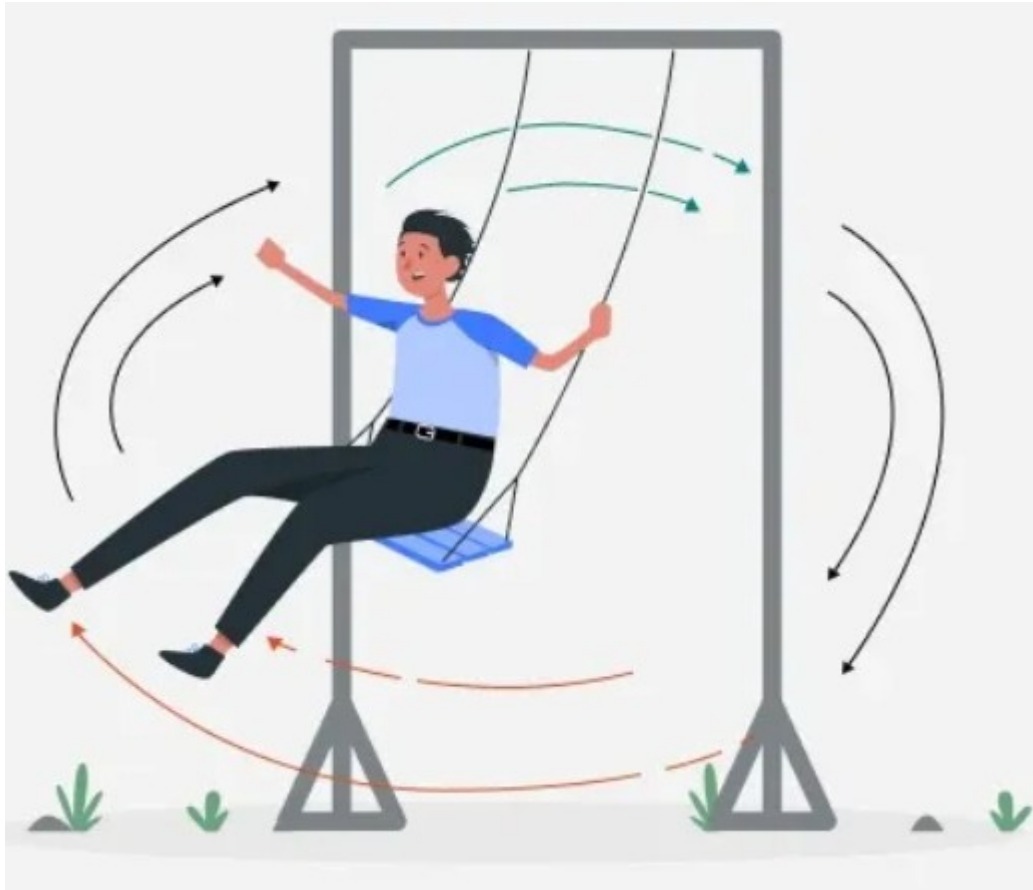
A matrix transformation may cause the vector to rotate (change direction – rotate, expand, contract) but in the case of eigenvector transformation, the vector's direction is unchanged or reversed, its magnitude changes depending upon the eigenvalue. Eigenvectors holds on to its direction steadfast or may reverse and change in magnitude.

Key Takeaway

Eigenvector = direction of the vector is preserved by transformation

Eigenvalue = scaling factor of the magnitude/length of the vector along that direction.

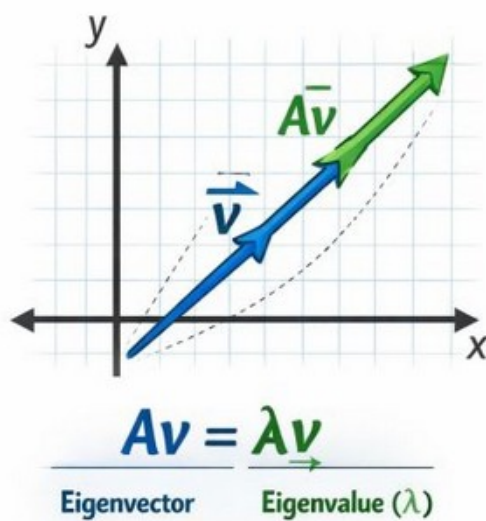
Illustration of eigenvector



Eigenvector: Direction of swing – same direction and reverses

Eigenvalue: High – more swing, low- less swing

<https://www.geeksforgeeks.org/engineering-mathematics/applications-of-eigenvalues-and-eigenvectors/>



Eigenvalues & Eigenvectors

Eigenvector

$\mathbf{v}$: A vector that remains in the same direction after a linear transformation.



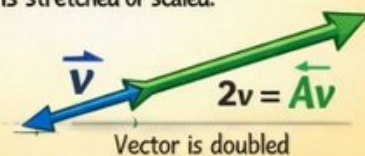
Stretched / Scaled

Eigenvalue

λ : The scalar factor by which the eigenvector is stretched or scaled.

E.g., $\lambda = 2$

Vector is doubled



Vector is doubled

https://www.linkedin.com/posts/sofienkaabar_eigenvalues-eigenvectors-and-what-they-activity-7419703259189256192-TA03/

5. Positive Definite Matrix (PDM)

This matrix enjoys the following properties

- All its eigenvalues are positive > 0
- The matrix is invertible
- It may be decomposed by Cholesky decomposition techniques – $LL' = \text{PDM}$
- The matrix is symmetric
- The matrix has a positive determinant
- For any non zero vector $\mathbf{u}$ we have $\mathbf{u}^T \text{PDM} \mathbf{u} > 0$ (quadratic form is always positive)
- Diagonal entries are positive

Example: Show that the following 3x3 matrix **A** is positive definite

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

% Features of positive definite matrix

```
PDM = [ 2 -1 0
        -1 2 -1
         0 -1 2];
[evePDM evaPDM] = eig(PDM);
determinant_PDM = det(PDM);
Symmetric_PDM = PDM';
```

% eigenvectors of PDM are
%

```
% evePDM =
%
% 0.5000 -0.7071 -0.5000
% 0.7071 0.0000 0.7071
% 0.5000 0.7071 -0.5000
```

% eigenvalues of PDM are

```
% evaPDM =
%
% 0.5858 0 0
% 0 2.0000 0
% 0 0 3.4142
```

% Determinant of PDM are
% det(PDM) = 4

% Symmetric PDM' = PDM

```
out_PDM = [PDM PDM'];
```

% out_PDM =

```
% 2 -1 0 2 -1 0
% -1 2 -1 -1 2 -1
% 0 -1 2 0 -1 2
%
```

% Both PDM and PDM' are the same

```

% Inverse of PDM exists
PDM_Inv = inv(PDM);

% PDM_Inv =
%
%   0.7500   0.5000   0.2500
%   0.5000   1.0000   0.5000
%   0.2500   0.5000   0.7500

% Note the inverse of PDM is also symmetric just like PDM itself

% u'PDMu>0;

u=[-1 -1 -1]';

uTPDMu = u'*PDM*u;

% uTPDMu = 2

% Diagonal entries are positive

diagonal_PDM = diag(diag(PDM));

% 1. Upper Triangular Factor (R)
R = chol(PDM);
% Result: R'*R equals PDM

% R =
%
%   1.4142  -0.7071    0
%    0    1.2247  -0.8165
%    0    0    1.1547
%
% R'*R =
%
%   2.0000  -1.0000    0
%  -1.0000   2.0000  -1.0000
%    0   -1.0000   2.0000

% Lower Triangular Factor (L)

L = chol(PDM, 'lower');

% L =
%
%   1.4142    0    0

```

```
% -0.7071  1.2247    0
%      0 -0.8165  1.1547
```

% Result: $L \cdot L'$ equals PDM

```
% L*L'
%
% ans =
%
%  2.0000 -1.0000    0
% -1.0000  2.0000 -1.0000
%      0 -1.0000  2.0000
```

6. Orthogonal Matrices (OGM)

A matrix is orthogonal if it has the following properties

a) Square matrix (equal number of rows and columns)

b) Matrix columns and rows are orthonormal

c) Transpose = Inverse i.e.,

$$\mathbf{OGM}' = \mathbf{OGM}^{-1}$$

Such that

$$\mathbf{OGM} \cdot \mathbf{OGM}' = \mathbf{I}$$

d) Columns and rows are having unit length

e) Dot product of different columns = 0

f) Dot product of different rows = 0

g) Determinant of **OGM** = -1 or +1

h) When you multiply two orthogonal matrices you get the resulting matrix also as orthogonal

i) Orthogonal matrices preserve the length of vectors. If we apply an orthogonal matrix **OGM** on vector **a** to get vector **b** then both **a** and **b** have the same length. Hence the vectors are neither contracted or expanded upon transformations using the orthogonal matrix.

Let

$$OGM^* a = b$$

Then

$$\|a\| = \|b\|$$

j) Angles between vectors are preserved on orthogonal transformation

Example of an orthogonal matrix

$$OGM_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Calculations demonstrating the properties of the orthogonal matrix are given in the next page

% Orthogonal Matrices

```
OGM1 = [1/sqrt(2) -1/sqrt(2)
        1/sqrt(2) 1/sqrt(2)];
```

% Square matrix (equal number of rows and columns)

```
% Order_OGM1 = size(OGM1);
```

%

% b) Matrix columns and rows are orthonormal meaning unit length

```
ONC1= norm(OGM1(:,1));
```

```
ONC2= norm(OGM1(:,2));
```

```
ONR1= norm(OGM1(1,:));
```

```
ONR2= norm(OGM1(2,:));
```

```
out_ON=[ONC1 ONC2 ONR1 ONR2];
```

```
% out_ON = 1.0000 1.0000 1.0000 1.0000
```

% c) Transpose = Inverse i.e.,

```
% OGM'=inv(OGM)
```

```
out_TI=[OGM1' inv(OGM1)];
```

```
%out_TI =
```

```
% 0.7071 0.7071 0.7071 0.7071
```

```
% -0.7071 0.7071 -0.7071 0.7071
```

% Such that

```
% OGM*OGM'=I
```

```
IsitI = OGM1*OGM1';
```

```
% IsitI = Is it identity matrix
```

```
%
```

```
% 1.0000 0
```

```
% 0 1.0000
```

% YES

% d) Columns and rows are having unit length

% Already checked above see b)

% e) Dot product of different rows = 1 if rows are same

```

% Dot product of different rows = 0 if rows are not same
% Dot product of different rows = 1 if rows are same

norows=numel(OGM1(:,1)); % number of elements in first column of OGM1

for i=1:norows % number of elements in first column of OGM1

    for j= 1:norows
        prod_rows(i,j) = (OGM1(i,:))*(OGM1(j,:))';

    end
end

% prod_rows =

% 1.0000    0
% 0    1.0000

% e) Dot product of different columns = 1 if columns are same
% Dot product of different columns = 0 if columns are not same

nocolumns=numel(OGM1(1,:));

for i=1:nocolumns % number of elements in first row of OGM1

    for j= 1:nocolumns

        prod_columns(i,j) = (OGM1(:,i))'*(OGM1(:,j));

    end
end

% prod_columns =
% 1.0000    0
% 0    1.0000

% f) Dot product of different rows = 0 See above
% g) Determinant of OGM = -1 or +1

Determinant_OGM1 =det(OGM1);

% Determinant_OGM1 = 1.0000

% h) When you multiply two orthogonal matrices you get the resulting matrix also
as orthogonal

```


% Let us multiply OGM1 with itself

```
OGM11 = OGM1*OGM1;
```

```
% 0 -1.0000
% 1.0000 0
```

```
IsitI2 = OGM11'*OGM11; % Is it identity matrix? YES
```

```
% IsitI2 =
```

```
%
% 1.0000 0
% 0 1.0000
```

% i) Orthogonal matrices preserve the length of vectors.

% If we apply an orthogonal matrix OGM on vector a to get vector b then both a and b have the same length.

% Hence the vectors are neither contracted or expanded upon transformations using the orthogonal matrix.

```
%
```

```
a=[-10 +20]';
length_a=norm(a);
b= OGM1*a;
length_b=norm(b);
```

```
% a =
```

```
%
% -10
% 20
%
```

```
% b =
```

```
%
% -21.2132
% 7.0711
%
```

```
%
```

```
%
```

```
% length_a = 22.3607
```

```
%
```

```
% length_b = 22.3607
```

```
% Lengths are preserved on orthogonal transformation
```

% Angles are preserved in orthogonal transformation

% Let OGM1 apply on any two vectors c and d represented row wise

```
c=[1 -4 7];
d=[-7 2 5];
```

```

% Finding angle between two vectors

% Angle in radians
theta = atan2(norm(cross(c,d)), dot(c,d));
% Matlab command to find angles between two vector
% Angle in degrees
theta_deg_cd = rad2deg(theta);

% The angles between vectors c and d is theta_deg = 73.8145°

% Let us now apply orthogonal transformation on vectors c and d
% simultaneously to get new two vectors c2 and d2

% Define a new orthogonal matrix OGM3 of size 3x3

OGM3 = [1/3 2/3 2/3; 2/3 1/3 -2/3; 2/3 -2/3 1/3];

% OGM3 =
%
%    0.3333    0.6667    0.6667
%    0.6667    0.3333   -0.6667
%    0.6667   -0.6667    0.3333

NewMatrix_cd = OGM3*[c' d'];

% NewMatrix_cd =
%
%    2.3333    2.3333
%   -5.3333   -7.3333
%    5.6667   -4.3333

c2 = NewMatrix_cd(:,1)'; % First vector from NewMatrix_cd
d2 = NewMatrix_cd(:,2)'; % Second vector of NewMatrix_cd

% c2 = 2.3333   -5.3333    5.6667
% d2 = 2.3333   -7.3333   -4.3333


% Angle in radians
theta = atan2(norm(cross(c2,d2)), dot(c2,d2));
% Angle in degrees
theta_deg_c2d2 = rad2deg(theta);
compare_angles = [theta_deg_cd  theta_deg_c2d2];

% compare_angles =    73.8145    73.8145

```

% Angles are preserved even after orthogonal transformation just as lengths
% :)